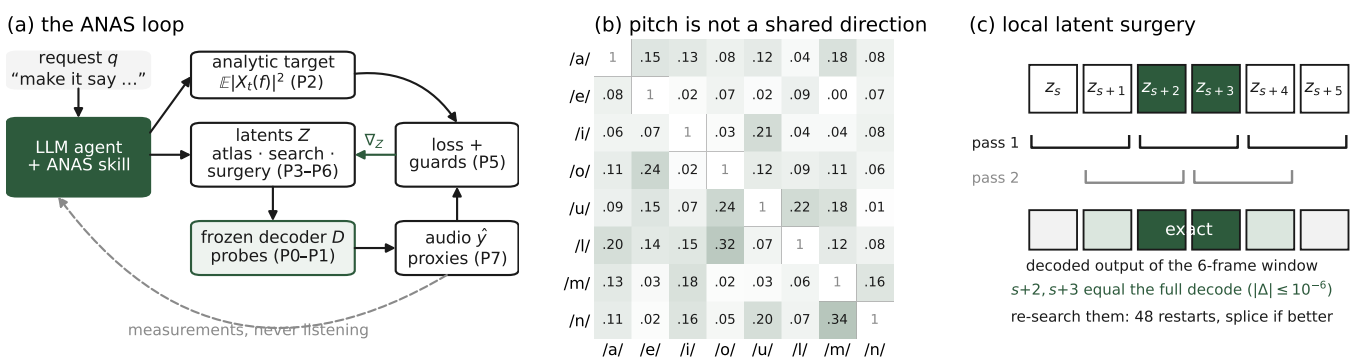
(a) the ANAS loop
request q
"make it say ..."
LLM agent
+ ANAS skill
analytic target
$\mathbb{E}|X_t(f)|^2$ (P2)
latents Z
atlas · search ·
surgery (P3–P6)
∇_Z
loss +
guards (P5)
frozen decoder D
probes (P0–P1)
audio $\hat{y}$
proxies (P7)
measurements, never listening
(b) pitch is not a shared direction
/a/ 1 .15 .13 .08 .12 .04 .18 .08
/e/ .08 1 .02 .07 .02 .09 .00 .07
/i/ .06 .07 1 .03 .21 .04 .04 .08
/o/ .11 .24 .02 1 .12 .09 .11 .06
/u/ .09 .15 .07 .24 1 .22 .18 .01
/l/ .20 .14 .15 .32 .07 1 .12 .08
/m/ .13 .03 .18 .02 .03 .06 1 .16
/n/ .11 .02 .16 .05 .20 .07 .34 1
/a/ /e/ /i/ /o/ /u/ /l/ /m/ /n/
|cos|, below diagonal: male (mean 0.12), above: female (0.09)
(c) local latent surgery
z_s z_{s+1} z_{s+2} z_{s+3} z_{s+4} z_{s+5}
pass 1
pass 2
exact
decoded output of the 6-frame window
$s{+}2, s{+}3$ equal the full decode ($|\Delta| \le 10^{-6}$)
re-search them: 48 restarts, splice if better